

features, communication interfaces, or application-specific requirements. Such customisation is often difficult to obtain from overseas suppliers, particularly for moderate production volumes.

This approach has helped RapidRadio evolve from a niche technology developer into a supplier serving multiple industries. Its RFID solutions are deployed in more than 500 libraries across India, reducing book issue and return times from nearly 20 seconds to less than five seconds. The company's technology is also used in pathology equipment, EV battery-swapping infrastructure, vehicle-access systems, parking automation platforms, industrial asset-tracking solutions, and retail applications. A notable deployment is with Transasia Biomedical, where RFID readers authenticate consumables used in diagnostic machines. RapidRadio also supplies technology to customers and partners associated with Tata Group companies, Godrej, Sun Mobility, and numerous RFID system integrators across India.

Unlike many electronics startups that outsource most engineering



Dhaval H. Kotecha, Managing Director, RapidRadio

activities, RapidRadio has retained its core design capabilities in-house. Manufacturing is outsourced to electronics manufacturing services (EMS) partners, while programming, calibration, and final quality checks remain under the company's direct supervision. This model enables RapidRadio to concentrate on innovation while continuing to leverage scalable manufacturing capacity.

The company has steadily converted technological expertise into commercial growth. It currently employs about 55 people and reported a provisional turnover of

₹142.5 million in the last financial year, during which it shipped approximately 11,000 RFID readers. Importantly, this growth has been driven largely by customer revenues. An early investment from Gujarat Venture Finance Limited in 2007 was fully repaid by 2013, and the company has continued to expand without relying on continuous external funding.

Looking ahead, RapidRadio is applying the same philosophy to contactless payments. It is developing indigenous payment modules that support RuPay and other payment ecosystems, allowing manufacturers to build point-of-sale terminals, EV charging payment systems, transit-ticketing infrastructure, ATM solutions, and other payment devices using locally developed electronics.

At a time when Make in India discussions often emphasise local assembly, RapidRadio represents a deeper level of technological self-reliance. By creating foundational hardware platforms that other manufacturers can build on, RapidRadio is reducing dependence on imported technology.

—Akanksha Sondhi Gaur

## 2 VERVESEMI INTEGRATED AI/ML-BASED ANALOGUE IP BLOCKS FOR CHIPS

An Indian startup is developing semiconductor IP blocks that enable chips to adapt to signal imperfections through real-time AI/ML-driven learning within silicon systems.

Most people think of a chip as one complete product. In reality, modern semiconductors are built from many smaller functional blocks. Vervesemi focuses on creating these critical internal building blocks, known as semiconductor intellectual property (IP), that other chipmakers can integrate into their own designs.

Founded in 2017 by Pratap Narayan Singh and Rakesh Malik, the company was created to develop original semiconductor technology in India rather than merely contributing engineering talent to overseas

firms. It operates as a fabless semiconductor company, concentrating on intellectual property, product design, and architecture, while outsourcing manufacturing to commercial semiconductor fabrication facilities. The name combines 'Verve,' meaning life, with 'Semi,' referring to semiconductors. Together, the two reflect the company's vision of bringing semiconductors to life.

What Vervesemi created first was not a 'chip' in the everyday sense. It was a semiconductor IP block, a pre-designed circuit module. In this context, an IP is not software, an app,

or a standalone product. Instead, it is a carefully engineered circuit design that can include both analogue and digital components. Once manufactured, it becomes a functional part of a larger semiconductor chip.

However, Vervesemi did something unusual with its analogue IP blocks. The company integrated a machine learning-based system directly at the silicon level, tightly coupled with the analogue circuitry. This system was not predictive in the abstract sense; it did not assume conditions before manufacturing. Instead, it operated in real